law of thermodynamics might permit one to make a perpetual motion machine, when it is a black hole that causes the violation, no perpetual motion machine is possible. The violation is just a tiny peculiarity in the laws of physics, one that the laws presumably live with quite happily.

Bekenstein was not convinced.

All the world's black-hole experts lined up on Hawking's side—all, that is, except Bekenstein's mentor, John Wheeler. "Your idea is just crazy enough that it might be right," Wheeler told Bekenstein. With this encouragement, Bekenstein plowed forward and tightened up his conjecture. He estimated just how much a hole's entropy would have to grow, when a package of air is dropped into it, to preserve the second law of thermodynamics, and he estimated how much the plunging package would increase the horizon's area; and from these rough estimates, he deduced a relationship between entropy and area which, he thought, *might* always preserve the second law of thermodynamics: The entropy, he concluded, is approximately the horizon's area divided by a famous area associated with the (as yet ill-understood) laws of quantum gravity, the *Planck–Wheeler area*, 2.61×10^{-66} square centimeter.[4] (We shall learn the significance of the Planck–Wheeler area in the next two chapters.) For a 10-solar-mass hole, this entropy would be the hole's area, 11,000 square kilometers, divided by the Planck—Wheeler area, 2.61×10^{-66} square centimeter, which is roughly 10^{79}.

This is an enormous amount of entropy. It represents a huge amount of randomness. Where does this randomness reside? Inside the hole, Bekenstein conjectured. The hole's interior must contain a huge number of atoms or molecules or something, all randomly distributed, and the total number of ways they could be distributed must be[5] $10^{10^{79}}$

Nonsense, responded most of the leading black-hole physicists, including Hawking and me. The hole's interior contains a singularity, not atoms or molecules.

Nevertheless, the similarity between the laws of thermodynamics and the properties of black holes was impressive.

In August 1972, with the golden age of black-hole research in full swing, the world's leading black-hole experts and about fifty students congregated